

Three Measures of Gravitational Identity

Budget, Jurisdiction, and Compression
in Weak-Field Curvature Attribution

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Abstract

A finite distribution of matter produces a finite contribution to the gravitational curvature of spacetime. In the Newtonian weak-field limit, this contribution admits three separable measures of gravitational identity: an intrinsic exterior Weyl-electric square budget, a relational jurisdiction bounded by source-context parity, and a tolerance-dependent compression boundary defined by multipole truncation fidelity. Each measure is independently computable and operationally distinct. The budget depends on mass and radius alone. The jurisdiction depends on the surrounding source scene. The compression depends on a declared approximation tolerance. For a declared spherical source, the exterior Weyl-electric square budget is $B_E = 8\pi G^2 M^2 / R^3$, which converges without cutoff. The budget per unit mass is algebraically proportional to mean density. Applied to declared spherical source models across astrophysical scales, the budget spans seventy-two orders of magnitude, peaks near the neutron star / black hole boundary under declared equations of state, and inverts for black holes, where $B_E = \pi c^6 / (GM)$. The construction introduces no new field equation, force, or modification of general relativity. It is a readability result within a declared weak-field regime: the gravitational identity of a finite source is structured, separable, and computable.

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1 Introduction

A finite body of matter curves the space around it. This much is old news, and belongs to Newton before it belongs to Einstein. What has received less attention is the structure of that curving—not its dynamics, not its field equations, but its *bookkeeping*. How much total curvature does a finite source produce? Over what region does that curvature dominate the surrounding gravitational environment? At what distance does the resolved detail of the source’s internal structure become irrelevant?

These are three different questions. They have three different answers. And those answers depend on different things. The first depends only on what the source is—its mass and its size. The second depends on what surrounds it—the masses and positions of neighboring sources. The third depends on what you intend to do with the information—the tolerance you declare for approximation.

This paper names these three answers, computes them for controlled models, demonstrates their independence, and deploys them across the full astrophysical scale hierarchy. The result is not a new theory of gravity. It is a diagnostic separation within standard weak-field gravitational physics, applied to the question of what it means for a finite source to exist gravitationally.

The three measures are:

Budget. The total exterior Weyl-electric square content of the source—a single number, finite by convergence, determined entirely by the source’s mass and radius. This is the gravitational fingerprint of the source’s existence: how much trace-free tidal curvature it writes into the surrounding geometry.

Jurisdiction. The region of space in which the source’s tidal contribution dominates over the combined contributions of its context. Bounded by the parity surface where source and context reach equality. This is relational: the same source in a different scene has a different jurisdiction.

Compression. The distance beyond which the source’s internal structure—its shape, density profile, higher multipole moments—can be replaced by a simpler representation at a declared tolerance. This is operational: it depends on how much fidelity the observer or computation demands.

In the spherical-source cases examined here, the ordering is consistent. Compression is innermost—detail washes out closest to the source. Budget extends everywhere but is concentrated near the source. Jurisdiction is outermost, bounded by context. The three surfaces do not coincide, cannot be derived from one another without additional input, and answer genuinely different questions about the same object.

Prior work in the Atlas program developed the mathematical machinery used here. The Gravitational Coherence Surface program (GCS I–VI) constructed source-context parity surfaces [1], many-source parity networks [2], mesoscale readability diagnostics [3], extraction and visualization methodology [4], nesting-domain handoff [5], and the Gravitational Field-Relation Operator (GFRO) [6]. A companion paper demonstrated boundary-supported Weyl-electric attribution for a single resolved body and the scalar-tensor channel divergence [7]. The present paper draws on all of these, but stands independently: the three-measure separation is the new result, and it requires only standard weak-field physics to state and demonstrate.

2 Scope and Non-Claims

This paper works in the Newtonian weak-field limit. The trace-free tidal tensor $E_{ij} = T_{ij} - \frac{1}{3}\delta_{ij}\text{tr}(T)$ serves as the static proxy for the electric part of the Weyl tensor. All sources are declared models with declared radii. All demonstrations use the convention $\nabla^2\Phi = 4\pi G\rho$, with the tidal tensor $T_{ij} = \partial_i\partial_j\Phi$.

The paper does not modify general relativity. It does not introduce a new force, field, source, or material boundary. It does not claim observability of the budget or the parity surface. It does not solve the dynamical handoff problem—how curvature attribution propagates when sources change—though it identifies that problem precisely. It does not claim that the particle entries in the budget ladder (proton, electron, etc.) are physical models of those particles. They are effective-source illustrations using declared masses and radii.

The budget is defined as a specific integral: the exterior Weyl-electric square budget, denoted B_E . It is not the Kretschmann scalar, though the two are related in vacuum Schwarzschild spacetime. The naming must match the regime: this is a Newtonian trace-free tidal-square integral, not a full general-relativistic curvature invariant.

3 Curvature Channels

In the Newtonian limit, the tidal tensor of a mass distribution separates into two channels. The trace carries the local density:

$$\text{tr}(T) = \nabla^2\Phi = 4\pi G\rho. \quad (1)$$

This is the Ricci-side channel. It records where mass-energy is locally registered as curvature. Inside a body, it is nonzero and proportional to the local density. Outside a body, it vanishes.

The trace-free part carries shape and anisotropy:

$$E_{ij} = T_{ij} - \frac{1}{3}\delta_{ij}\text{tr}(T). \quad (2)$$

This is the Weyl-electric channel. It records how the source’s geometry—its oblateness, its internal density gradients, its finite extent—is expressed as trace-free tidal structure. Inside a uniform sphere, it vanishes. Inside an oblate body, it does not. Outside any body, it is the entire tidal tensor.

These two channels answer different bookkeeping questions. The Ricci-side channel asks: where is mass-energy locally accounted for? The Weyl-electric channel asks: how is the source’s shape and structure expressed as tidal geometry? A scalar diagnostic (such as $Q = GM/r^3$) compresses both channels into a single number. The tensor diagnostic preserves their separation. This distinction is not merely technical—it determines whether a given partition of a source scene produces an equality surface or not, as demonstrated in the companion paper [7].

4 The Three Measures

4.1 Budget: What You Are

The exterior Weyl-electric square budget of a spherical source with mass M and radius R is defined as

$$B_E = \int_R^\infty E_{ij} E^{ij} \cdot 4\pi r^2 dr. \quad (3)$$

For a spherical source, the exterior trace-free tidal tensor has the eigenvalue structure $(-GM/r^3, -GM/r^3, 2GM/r^3)$, giving

$$E_{ij} E^{ij} = \frac{6 G^2 M^2}{r^6}. \quad (4)$$

The integral evaluates to

$$\boxed{B_E = \frac{8\pi G^2 M^2}{R^3}}. \quad (5)$$

This integral converges without any external cutoff. The $1/r^6$ decay of the quadratic curvature invariant exceeds the r^2 growth of the spherical volume element, producing a finite result from an integral over infinite extent. No parity surface is needed. No context is invoked. The budget is intrinsic.

The budget is concentrated near the source. The fraction of B_E contained within radius $r > R$ is

$$f(r) = 1 - \left(\frac{R}{r}\right)^3. \quad (6)$$

Thus 87.5% lies within $2R$, 99.2% within $5R$, and 99.9% within $10R$.

The budget per unit mass admits an exact algebraic identity:

$$\frac{B_E}{M} = \frac{32\pi^2}{3} G^2 \bar{\rho}, \quad (7)$$

where $\bar{\rho} = 3M/(4\pi R^3)$ is the mean density. Budget efficiency — curvature produced per unit mass — is proportional to density and to nothing else.

4.2 Jurisdiction: Where You Matter

The jurisdiction of a source A in a scene of sources is the region where its Weyl-electric tidal contribution dominates over the combined contribution of the context K :

$$\text{Jurisdiction}(A) = \{x : \|E^{(A)}(x)\|_F > \|E^{(K)}(x)\|_F\}. \quad (8)$$

The boundary is the parity surface, the zero set of the GFRO Frobenius residual [6]:

$$\Sigma_R[A|K] = \{x : \|E^{(A)}(x)\|_F = \|E^{(K)}(x)\|_F\}. \quad (9)$$

At the parity surface, the attribution entropy between source and context is maximized. Beyond it, the source's field is still present but no longer dominant.

Jurisdiction is relational. As illustrative R0-based estimates under point-source approximation: the same proton in a hydrogen atom would hold jurisdiction over roughly 49,000 fm; the same proton in a deuteron, approximately 1 fm. These are effective-model

illustrations, not derived Weyl-electric results. Same source, same budget, different jurisdiction.

Jurisdiction is finite when the parity surface closes, which requires that context sources provide sufficient competing contribution. In physical scenes with declared context, the parity surface generically closes and the jurisdiction is bounded. The budget is intrinsically finite. The jurisdiction is contextually finite, contingent on the existence of a closed parity surface. These are different kinds of finiteness with different origins.

4.3 Compression: When Your Detail Fades

The compression boundary of a source is the radius beyond which its resolved internal structure can be replaced by a truncated multipole representation at a declared tolerance τ :

$$r_\tau(L_{\max}) = \inf \left\{ r : \max_{\theta} \frac{|\Phi_{L_{\max}}(r, \theta) - \Phi_{\text{ref}}(r, \theta)|}{|\Phi_{\text{ref}}(r, \theta)|} \leq \tau \quad \forall r' \geq r \right\}. \quad (10)$$

This is an approximation-validity boundary. For a uniform sphere, all higher multipole moments vanish and the compression boundary is the body surface itself. For an oblate body, the compression radii contract inward as more multipole terms are retained [7].

Compression depends on tolerance, truncation order, and field quantity. It is not a parity surface and not a budget.

4.4 Independence

The three measures are operationally independent. Each can change while the others remain fixed. A source can have its budget change (by changing mass or density) while jurisdiction and compression hold constant. A source can have its jurisdiction change (by changing context) while budget and compression hold constant. A source can have its compression change (by changing tolerance) while budget and jurisdiction hold constant.

No two of the three can be derived from the third without additional input. Budget requires mass and radius. Jurisdiction requires the source scene. Compression requires a tolerance declaration. They are three independent coordinates of gravitational identity.

5 Demonstration: Single Resolved Body

The companion paper [7] develops the boundary-supported Weyl-electric attribution construction for a uniform-density oblate spheroid with $a = 1$, $c = 0.5$, $\rho_0 = G = 1$. The key results, verified by independent reproduction:

The boundary reconstruction recovers the analytic interior tidal tensor to machine precision ($\sim 10^{-16}$ relative Frobenius error). Symmetric boundary partitions recover the expected parity planes. Asymmetric partitions (polar cap versus equatorial belt) reveal the scalar-tensor channel divergence: the scalar Q-based readable produces an interior equality surface while the Weyl-electric Frobenius readable does not, for the same partition. Exterior multipole compression radii contract inward with increasing truncation order.

These results validate the computational tools used throughout the present paper.

6 Demonstration: Two Resolved Bodies

To demonstrate the jurisdiction measure, we construct a scene with two resolved oblate spheroids: Body A ($a = 1.0$, $c = 0.5$, $\rho = 1.0$, $M = 2.094$) centered at the origin, and Body B ($a = 0.7$, $c = 0.35$, $\rho = 1.5$, $M = 1.078$) centered at $(4, 0, 0)$.

Each body's Weyl-electric tidal tensor is computed at every scene point via its own boundary integral. Tensor closure holds to $\sim 10^{-15}$: $E_{ij}^{(A)} + E_{ij}^{(B)} = E_{ij}^{(\text{total})}$ everywhere.

6.1 Parity surface

The Weyl-electric Frobenius parity surface forms between the two bodies, crossing the connecting axis at $x \approx 2.228$. The distance ratio $d_A/d_B \approx 1.258$ is nearly constant across the full angular extent, indicating a near-Apollonius geometry. The tensor correction from the scalar R0 prediction $((M_A/M_B)^{1/3} \approx 1.247)$ is approximately 0.01 in the ratio.

6.2 Budget and compression at parity

Both bodies' quadrupole compression radii enclose the parity surface. The spatial ordering along the connecting axis is:

A surface $\rightarrow$ A octupole compression $\rightarrow$ B quadrupole compression $\rightarrow$ **Parity** $\rightarrow$
A quadrupole compression $\rightarrow$ B surface

Resolved internal structure matters at the jurisdictional boundary. The tensor-readable geometry reads the bodies' shapes, not just their masses.

6.3 The context bath

Inside Body A, Body B's tidal field is small but spatially varying: 1.4% of the self-field at the far interior, growing to 2.8% at the near surface. Body A's own interior field is spatially constant (the uniform-ellipsoid result). The self-field is isotropic in magnitude; the context field has direction. The peer's tidal curvature permeates the body's interior with a gradient pointing toward the neighbor.

7 The Budget Ladder

The budget formula $B_E = 8\pi G^2 M^2 / R^3$ applies to any declared spherical source model. Applied across the full scale hierarchy, it produces the ladder in Table 1.

Table 1: Exterior Weyl-electric square budget across astrophysical scales. Particle entries use declared effective radii and are illustrative.

Object	M (kg)	R (m)	B_E (m ³ /s ⁴)	$\log_{10} B_E$
electron	9.11×10^{-31}	3.86×10^{-13}	1.61×10^{-42}	-41.8
proton	1.67×10^{-27}	8.75×10^{-16}	4.68×10^{-28}	-27.3
Fe-56 nucleus	9.30×10^{-26}	3.74×10^{-15}	1.85×10^{-26}	-25.7
hydrogen atom	1.67×10^{-27}	5.29×10^{-11}	2.12×10^{-42}	-41.7
Earth	5.97×10^{24}	6.37×10^6	1.54×10^{10}	10.2
Sun	1.99×10^{30}	6.96×10^8	1.32×10^{15}	15.1
white dwarf	1.19×10^{30}	6.96×10^6	4.74×10^{20}	20.7
NS ($1.4 M_\odot$)	2.78×10^{30}	1.10×10^4	6.52×10^{29}	29.8
NS max ($2.2 M_\odot$)	4.38×10^{30}	1.05×10^4	1.85×10^{30}	30.3
BH ($10 M_\odot$)	1.99×10^{31}	2.95×10^4	1.72×10^{30}	30.2
Sgr A*	7.96×10^{36}	1.18×10^{10}	4.30×10^{24}	24.6
TON 618	1.31×10^{41}	1.95×10^{14}	2.60×10^{20}	20.4

7.1 The nuclear plateau

Protons and neutron stars have nearly identical budget-per-mass values ($B_E/M \approx 0.2$ – 0.4) despite differing in mass by a factor of 10^{57} . This is because they share nuclear density ($\sim 5 \times 10^{17}$ kg/m³). The curvature efficiency of nuclear matter is scale-invariant.

7.2 The atom cliff

The hydrogen atom has the same mass as the proton but a radius sixty thousand times larger. The budget drops fourteen orders of magnitude. Atomic structure—electron orbitals, chemical bonds, molecular geometry—is gravitationally invisible. The curvature budget of ordinary matter resides almost entirely in its nuclei.

7.3 The black hole inversion

For a Schwarzschild black hole with $R = r_s = 2GM/c^2$, the budget becomes

$$B_E^{\text{BH}} = \frac{\pi c^6}{G M}. \quad (11)$$

The budget is inversely proportional to mass. The most massive objects in the universe are, by this measure, the most curvature-dilute. TON 618—sixty-six billion solar masses—has a curvature budget comparable to a white dwarf.

7.4 The peak

The budget peaks near the neutron star / black hole boundary. Under a declared equation of state, a neutron star at the TOV limit has

$$B_{\text{max}} = \frac{8\pi G^2 M_{\text{TOV}}^2}{R_{\text{TOV}}^3}. \quad (12)$$

Within the declared spherical-source model and for stable matter obeying a given equation of state, this represents the maximum exterior Weyl-electric square budget. It is determined by the intersection of nuclear physics (the equation of state) and general relativity (the TOV equation). Whether this bound extends beyond the declared model class is not claimed here.

8 The Separation

The three measures are not three views of one quantity. They are genuinely independent.

A proton in a deuteron has the same budget as a proton in a hydrogen atom, but its jurisdiction is fifty thousand times smaller.

A sphere and an oblate body of the same mass have the same exterior budget but different compression boundaries, because the oblate body has nonzero quadrupole and higher moments.

Two identical protons in different crystalline environments have the same budget and the same compression, but different jurisdictions, because their neighbors differ.

Budget tells you what the source is. Jurisdiction tells you where the source matters. Compression tells you when the source's detail fades. These are three chapters of a gravitational biography, and they must be read separately.

9 The Handoff Problem

The three measures described in this paper are static. They describe a solved geometry on one time slice. The unsolved problem is dynamical.

If a source's mass distribution changes, its Weyl-electric field must update. That update propagates at the speed of light. The parity surfaces renegotiate. The jurisdiction boundaries shift. The compression radii adjust. The entire gravitational identity of the source undergoes a causal transformation that cannot be captured by comparing two static slices.

The contracted Bianchi identity provides a differential constraint: $\nabla^\mu C_{\mu\nu\rho\sigma} = 2\nabla_{[\rho} P_{\sigma]\nu}$. This gives a chain of custody — stress-energy to Ricci to Schouten gradients to Weyl divergence — but not a standalone evolution equation for the Weyl tensor. The missing object is the causal transfer function: how Ricci-side source custody becomes Weyl-electric field expression, and how that expression propagates at finite speed.

The static construction exposes this problem with precision. It shows that the accounts separate and close on each slice. It does not show how the accounts update between slices. The budget is finite on every slice. The jurisdiction is bounded on every slice. The compression is well-defined on every slice. But the causal geometry of curvature expression — the movie — remains the frontier.

10 Limitations

The budget formula $B_E = 8\pi G^2 M^2 / R^3$ assumes a spherical exterior. For significantly aspherical sources, the angular dependence of $E_{ij} E^{ij}$ would modify the integral. The correction is computable but not computed here.

The budget is defined for the exterior only. Interior contributions (nonzero for aspherical bodies) are not included in B_E . A total budget including interior trace-free tidal structure would require the full boundary-supported reconstruction of the companion paper [7].

All particle entries use declared effective radii. The proton charge radius is a measured quantity; the electron reduced Compton wavelength is an illustrative scale. Quark entries are quarantined as placeholder illustrations.

The dynamical handoff is exposed but not solved. The three measures are static diagnostics, not dynamical evolution laws.

11 Conclusion

A finite source carries three separable measures of gravitational identity. The budget records how much trace-free tidal curvature the source writes into the geometry. The jurisdiction records where that curvature dominates over context. The compression records when the source’s resolved detail can be replaced by a simpler representation.

The budget converges without cutoff. The jurisdiction is bounded by parity. The compression is bounded by tolerance. Three different kinds of finiteness, three different dependencies, one source.

The budget peaks at the neutron star boundary and inverts for black holes. The most massive objects in the universe are the most curvature-dilute. The budget per unit mass is algebraically proportional to mean density: nuclear matter carries the same curvature efficiency at one femtometer and at ten kilometers.

The field is seamless. The metric does not tear at the surface of a star, the edge of a nucleus, or the boundary of a proton. But curvature identity is structured. A finite body produces a finite, computable curvature budget, holds a finite, context-dependent jurisdiction, and compresses into simpler representations at finite, tolerance-dependent distances. These three quantities do not coincide, do not derive from one another, and do not collapse into a single number.

What you are, where you matter, and when your detail fades: these are the three measures of gravitational identity. They were always present in the weak-field curvature of finite sources. This paper names them, separates them, and computes them.

References

- [1] D. Scieszinski, “The Gravitational Coherence Surface,” GCS I, Relational Labs, 2025.
- [2] D. Scieszinski, “Many-Source Parity Networks,” GCS II, Relational Labs, 2025.
- [3] D. Scieszinski, “Mesoscale Gravitational Readability,” GCS III, Relational Labs, 2025.
- [4] D. Scieszinski, “Proof of Concept Extraction,” GCS IV, Relational Labs, 2025.
- [5] D. Scieszinski, “Nesting-Domain Handoff,” GCS V, Relational Labs, 2025–2026.
- [6] D. Scieszinski, “The Gravitational Field-Relation Operator: Direct Construction of Gravitational Parity Networks from Source-Context Residuals,” GCS VI, Relational Labs, 2026.

- [7] D. Scieszinski, “Boundary-Supported Weyl-Electric Attribution in a Uniform-Density Oblate Spheroid,” Relational Labs, 2026.